

MAYA STERLING

San Francisco, CA • (555) 010-9988 • maya.sterling.math@email.com • github.com/m-sterling-math

Professional Summary

Detail-oriented IT Professional transitioning from a strong academic background in Pure and Applied Mathematics. Expert in leveraging algorithmic logic, statistical analysis, and quantitative problem-solving to address complex technical challenges. Proficient in Python and SQL, with a deep understanding of the mathematical foundations of cryptography and data structures.

Technical Skills

Programming & Scripting: Python (NumPy, Pandas, Matplotlib), SQL, Bash Shell Scripting, Basic Java

IT Operations: Linux System Administration (Ubuntu/Debian), Git/GitHub, Docker Fundamentals

Mathematical Core: Discrete Mathematics, Linear Algebra, Multivariable Calculus, Number Theory, Probability & Statistics

Tools: Excel (Advanced Formulas/VBA), MATLAB, LaTeX, Jupyter Notebooks

Education

B.S. in Mathematics (Minor in Computer Science)

2020 – 2024

University of California, Berkeley • GPA: 3.85/4.00

- Recipient of the Excellence in Undergraduate Mathematics Award.
- Key Coursework: Abstract Algebra, Cryptography, Numerical Analysis, Data Structures, Operating Systems.

IT & Quantitative Projects

Network Optimization Engine (Python)

Applied *Graph Theory* and *Dijkstra's Algorithm* to simulate and optimize data packet routing across a mock network infrastructure.

- Reduced simulated latency by 22% through the implementation of efficient path-finding logic.
- Visualized network bottlenecks using Matplotlib to assist in capacity planning.

Cryptographic Data Vault

Developed a file encryption utility based on *RSA Algorithm* principles, utilizing *Modular Arithmetic* and *Prime Factorization* logic.

- Engineered a secure key generation script in Python to handle 2048-bit encryption keys.
- Documented technical specifications focusing on the mathematical proofs ensuring data integrity.

Experience

Quantitative Analysis Intern

2023 – 2024

FinTech Solutions Group

- Cleaned and structured datasets of 1M+ records using Python and SQL, ensuring 99% data accuracy for modeling.

- Automated weekly reporting processes using Python scripts, reducing manual workload by 15 hours/week.
- Translated complex statistical findings into actionable technical requirements for the software engineering team.

Certifications

- Google IT Support Professional Certificate (Coursera)
- CompTIA A+ Certification (Candidate)
- Python for Data Science and AI (IBM)